Mathematical Foundations of Image Synthesis

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July 20, 2026

1 Introduction

Image synthesis is the process of generating images from mathematical and algo rithmic descriptions. Modern graphics systems rely on **precise mathematical models** and efficient computation. *Even small numerical errors* can result in visible artifacts in rendered images.

Typical formulations involve scalar values such as x, indexed parameters like I_{ij}, and exponential terms e^{x^2}. These elements are combined to simulate realistic lighting behavior.

2 Core Components

2.1 Rendering Pipeline

The rendering pipeline consists of the following stages:

- Scene Definition
- Camera Configuration
- Light Placement

Important Shading Model

  - Diffuse Shading
  - Specular Highlights

2.2 Execution Order

1. Input Processing
2. Ray Construction
   (a) Primary Rays
   (a) Secondary Rays

3. Intersection Evaluation
    - Object Hit Test

3 Mathematical Modeling

Light interaction with surfaces can be expressed using the following equations.

$$L = I_{ij}.\cos(\theta) + R^2 \tag{1}$$

$$k = \frac{ab}{cd}$$

$$\begin{aligned} f(x) &= x^2 + 2x + 1 \\ &= (x+1)^2 \end{aligned} \tag{2}$$

$$f(x) = \begin{cases} y & \text{if } y \geq 0 \\ -x & \text{if } y < 0 \end{cases}$$

$$p(t) = \begin{cases} 0, t < 0 \\ e^{-t}, 0 <= t < 3, \\ \ln(t-2), t >= 3 \end{cases}$$

$$\mathcal{N}(x_i, \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{x_i - \mu}{2\sigma^2}}$$

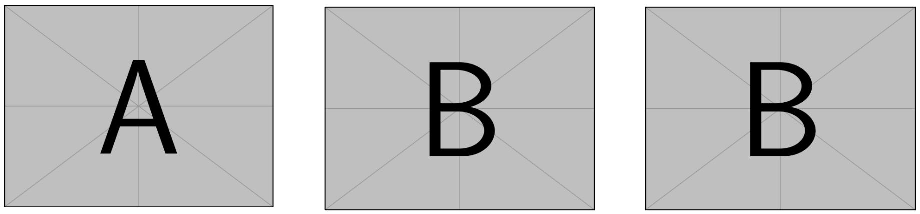
4 Performance Comparison

<table>
<tr><td>Method</td><td>Accuracy</td><td>Time Complexity</td></tr>
<tr><td>Rasterization</td><td>Medium</td><td>$O(n)$</td></tr>
<tr><td rowspan="2">Ray Tracing</td><td>High</td><td>$O(n^2)$</td></tr>
<tr><td>Very High</td><td>$O(n^3)$</td></tr>
<tr><td colspan="2">Hybrid Method</td><td>$O(n^2)$</td></tr>
</table>

Table 1: Comparison of Image Synthesis Techniques

The results in 1 highlight the trade-off between accuracy and computational cost.

5 Visual Illustration



(a) Output A (b) Output B (c) Output B

Figure 1: Caption

Figure 1 compares three different rendering outputs arranged in a single row.

6 Conclusion

Image synthesis combines mathematical precision, algorithmic design, and efficient implementation. As scene complexity increases, choosing the appropriate method becomes critical for balancing quality and performance.